「Norm.

「Prop. Let $f \in C([0,1])$. Then,

$$\lim_{n\to\infty}\left[\int_0^1 |f(x)|^n\,dx\right]^{\frac{1}{n}} = \max\{|f(x)| \mid x\in[0,1]\}.$$

Proof. Since $[0,1]$ is compact, f is continuous uniformly on $[0,1]$.
Given $\varepsilon>0$, choose $\delta>0$ such that $|x-y|<\delta \Rightarrow |f(x)-f(y)|<$
And, since $[0,1]$ is compact, the image of $|f|$ is also compact, so for some $a\in[0,1]$, $|f(a)|=\max\{|f(x)| \mid x\in[0,1]\}$. Observe that

$$|f(a)| = (|f(a)|^n)^{\frac{1}{n}} = \left[\int_0^1 |f(a)|^n\,dx\right]^{\frac{1}{n}}$$

Therefore, for each $n\in\mathbb{N}$,

$$\left|\left[\int_0^1 |f(x)|^n\,dx\right]^{\frac{1}{n}} - \left[\int_0^1 |f(a)|^n\,dx\right]^{\frac{1}{n}}\right|$$